MODEL DEVELOPMENT / EXPERIMENTAL RESULTS / NOVEMBER 2026

From Baseline to User-Aware Classification

An evidence-based retrospective of four Mini-BERT experiments on five-class Amazon electronics review ratings.

4	419,472	72.20%	72.06%
TRAINED VERSIONS	SELECTED LABELED REVIEWS	BEST VALIDATION MICRO-F1	FINAL INTERNAL TEST MICRO-F1

Executive assessment

The decisive improvement came from the richer text representation in V2. Increasing the input budget to 256 tokens, retaining the beginning and end of lengthy reviews, and introducing the review summary when needed raised validation Micro-F1 from **70.54% to 72.19%**. These changes were introduced together with a larger batch, so their individual contributions cannot be isolated from this run alone.

V3 tested a conservative stopword-removal policy, reducing token usage but also reducing predictive performance. V4 added temporally valid prior-rating information and reached the highest validation Micro-F1 (**72.203%**), only **0.0127 percentage points** above V2. Its learned history contribution was nearly zero on average.

Final result and test identity

The saved V4 model scored **72.057% Micro-F1** and **0.3680 MAE** on a **62,921-row internal held-out test**. This is not a score on the separate, unlabelled 20,000-row competition test file.

Experiment trajectory

V1 Baseline	V2 Enrich text	V3 Remove stopwords	V4 Add user history
128-token review-only	256-token conditional summary and head-tail	Conservative lexical filtering on V2	Temporal prior-rating features fused with V2

Source basis: six supplied executed notebooks, including their recorded figures, tables and predictions. All numeric claims below are grounded in those outputs; no retraining was performed for this report.

CHAPTER 01 / EXPERIMENTAL REVIEW

Experimental design and the V1 baseline

Research objective

Predict the **overall rating (1-5 stars)** from Amazon electronics review text, then determine whether improvements in text handling, preprocessing or user context produce genuine validation gains. Evaluation emphasizes Micro-F1, Macro-F1, per-class F1 and mean absolute rating error (MAE).

Controlled sampling protocol

Population / setup	Recorded configuration
Original labelled reviews	838,944 valid reviews
Selection strategy	Stratified random 50% subset, seed 42
Selected subset	419,472 reviews
Development partition	Train: 293,630 (70%) | Validation: 62,921 (15%)
Held-out internal test	62,921 (15%); excluded from model selection
Compute / model family	Google Colab Tesla T4; BERT Mini, uncased

Why half the data?

The initial goal was iteration speed, not an estimate of performance on all available training examples. A stratified half-sample preserved rating proportions while making repeated full fine-tuning practical on a T4. The trade-off is reduced training evidence and, later, less available historical behavior for each user.

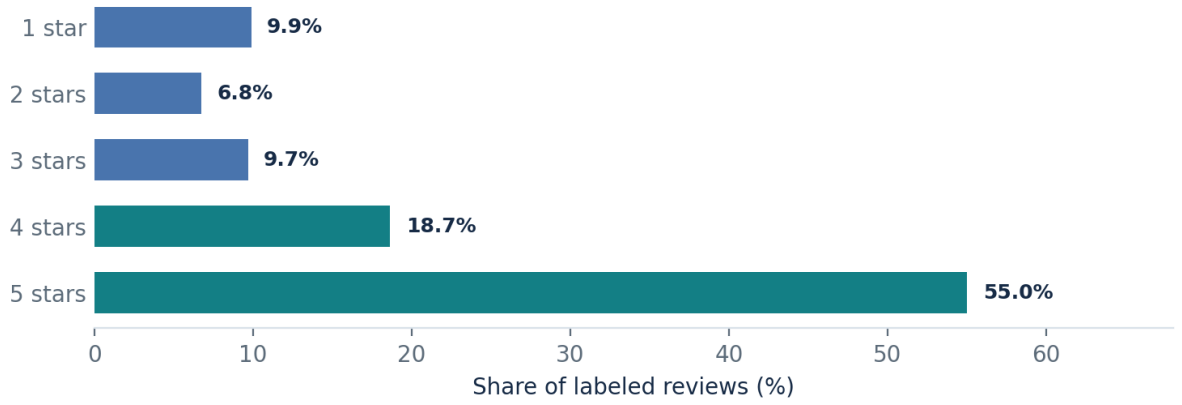


Figure 1. The target is imbalanced: five-star reviews account for about 55.0% of labeled examples. This makes per-class and Macro-F1 indispensable alongside Micro-F1.

V1: intentionally modest but fully fine-tuned

V1 used **google/bert_uncased_L-4_H-256_A-4** (4 Transformer layers, 256-dimensional hidden representations, 4 attention heads, approximately **11.17 million** parameters). A new five-class classification head was trained jointly with all pretrained parameters. Inputs consisted **only of reviewText**, truncated to **128 tokens including special tokens**; no summary, user ID or user history was used. Dynamic per-batch padding retained computational efficiency.

Recorded execution: batch size **64**, learning rate 2e-5, four epochs, best epoch 4. Validation Micro-F1 **70.539%**, Macro-F1 **54.706%**, MAE **0.4122**; training time **33.7 min**. This established the reference point against which later changes were assessed.

Source: Quera_V1(3).ipynb - configuration, split counts, epoch log and validation outputs. The executed batch size (64) takes precedence over a conflicting explanatory note in the notebook.

CHAPTER 02 / EXPERIMENTAL REVIEW

V2 - increasing useful text coverage

Decision and hypothesis

The 128-token baseline risked losing important clues in long reviews. V2 increased the context window to **256 tokens** and introduced a targeted input policy. Its objective was to preserve sentiment-bearing information both at the beginning and at the end, while exploiting the review summary when truncation became unavoidable.

Review length	Representation passed to BERT Mini
Fits in 256 tokens	Only the full reviewText; summary is ignored
Exceeds 256 tokens	Tokenize summary followed by reviewText; retain the first 192 and last 64 tokens, counting special tokens
Throughout	Dynamic batch padding; same 5-class fine-tuned architecture
Training batch	128 (versus 64 in V1); four epochs

Why this is one experiment, not three ablations

Maximum length, conditional use of summary / head-tail preservation, and training batch size all changed from V1 to V2. Their combined effect is measured; none can be assigned an isolated causal improvement from these results alone.

Observed outcome

Validation metric	V1	V2	V2 minus V1
Micro-F1 / accuracy	70.539%	72.191%	+1.651 pp
Macro-F1	54.706%	57.235%	+2.529 pp
MAE (lower is better)	0.4122	0.3701	-0.0421
Training time	33.68 min	20.12 min	-40.3%

Interpretation

V2 improves both overall and class-balanced performance, and makes fewer ordinal rating errors. The growth in Macro-F1 indicates that benefits extend beyond the dominant five-star class. The larger batch more than offsets the additional text processing in the measured wall-clock run; however, batch size and input length both changed, so the speedup cannot be attributed to either separately.

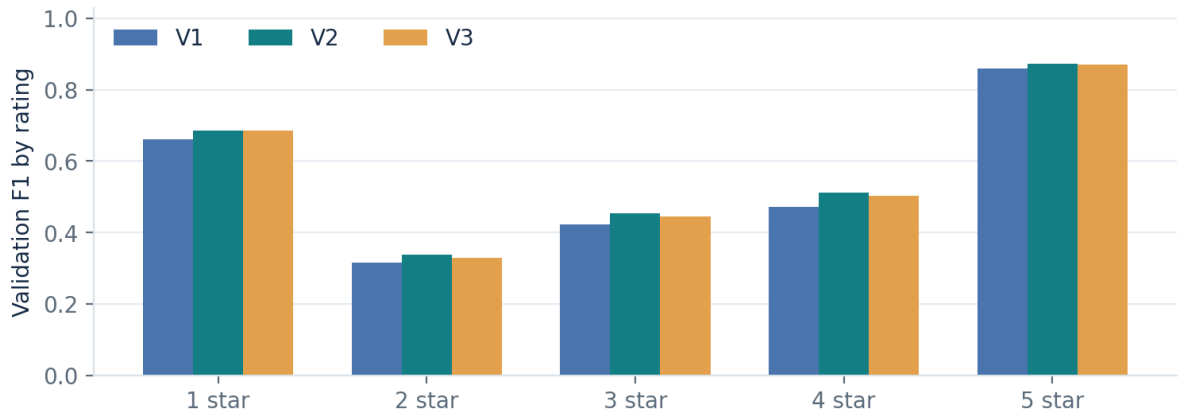


Figure 2. Validation F1 across rating classes. The most substantial practical difficulty remains predicting intermediate scores, especially two-star reviews.

CHAPTER 03 / EXPERIMENTAL REVIEW

V3 - testing conservative stopword removal

Reason for the experiment

With a 256-token budget, common function words appeared to consume space that might otherwise hold sentiment-rich content. V3 tested whether carefully removing a small, predefined group of articles, demonstratives and forms of “to be” could reduce truncation and improve classification.

Treatment and comparison

V3 preserved the V2 model, conditional summary / head-tail strategy, **256-token maximum**, **128 training batch** and four-epoch schedule. Its single proposed new intervention was rule-based removal from a 15-word candidate set. Negations, contractions, intensifiers, sensitive phrases, quotation spans, punctuation and identifiers were protected as far as possible. Decisions about whether a review counted as “long” were based on its original text, keeping the V2 entry condition consistent.

Token-use diagnostic (train)	Observed
Raw source-token reduction	5.66%
Actual model-input-token reduction	4.21%
Reviews needing truncation	12.24% to 11.70%
Newly fitting without truncation	1,577 train reviews
Preprocessing duration	8.54 minutes

Result: computationally leaner, statistically weaker

Validation metric	V2	V3	Change
Micro-F1	72.191%	71.814%	-0.377 pp
Macro-F1	57.235%	56.688%	-0.547 pp
MAE	0.3701	0.3765	+0.0064
Training time	20.12 min	20.42 min	+0.30 min

Analytical conclusion

Token saving did not translate into greater predictive quality. Language-model pretraining already exploits syntactic and functional context, and seemingly removable words can affect sentiment interpretation. V3 underperformed V2 on Micro-F1, Macro-F1 and MAE; consequently, the next iteration returned to the unfiltered V2 text policy. The extra preprocessing also removes the argument that this experiment clearly lowered end-to-end cost.

Source: Quera_V3(3).ipynb; logged token statistics and full validation metrics.

CHAPTER 04 / EXPERIMENTAL REVIEW

Feasibility check - can user history help?

Why inspect the original files first?

User context could be predictive only if the target test reviews include known users and if prior ratings are legitimately observable **before** the review being predicted. The investigation therefore preceded V4 training and examined the original full labelled train file against the separate unlabelled competition test file - without reading or requiring competition-test ratings.

Original-file audit	Recorded finding
Original labeled train	838,944 reviews; 321,135 unique users
External blind competition test	20,000 reviews; 19,037 unique users; no overall label
Shared user identities	15,541 (81.64% of test users)
Blind-test reviews from seen users	82.24% of test rows
Rows with strictly earlier rated train history	10,043 / 20,000 (50.215%)
Seen users but no earlier rated train review	6,405 (32.025%)
Unseen user rows	3,552 (17.760%)
At least 2 eligible prior ratings	6,246 (31.23% of test rows)

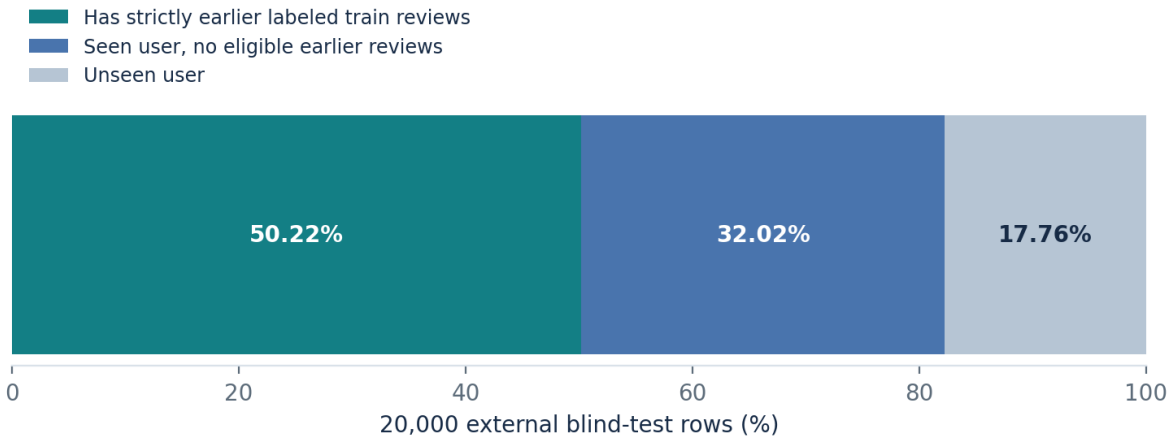


Figure 3. External competition test: whether strictly earlier labelled training history exists for each target review. Chronology matters more than identity overlap alone.

Decision following the audit

Feasibility supported: roughly half of the blind-test rows have at least one eligible earlier rating in the complete original training file. This motivated a history-aware model based on a user's prior rating mean and the number of earlier ratings. Crucially, 82.24% user overlap is **not** 82.24% usable history: 32.03% of test rows have a seen user but no strictly earlier train rating.

Time-safe definition

For a review on day D, history uses ratings by the same user on days **strictly before D** and only from permissible training data. Reviews on the same day are excluded, and no held-out target labels are used.

Source: test_train_user_comparison.ipynb; full original train/test audit. Coverage shown here uses the full 838,944-row labelled train file, not the 50% modelling subset.

CHAPTER 05 / EXPERIMENTAL REVIEW

V4 - using history as calibrated auxiliary evidence

Model strategy

V4 returned to the **V2 text formulation** (256 tokens, conditional summary, head-tail retention, batch 128) and added two historical predictors: **the mean of a user's previous ratings** and **the count of eligible previous ratings**. All history was reconstructed from earlier training observations. Train, validation and internal-test membership remained on the same controlled split as earlier versions.

History fusion, expressed conceptually

A lightweight learned history branch predicts a five-star probability distribution. A trainable gate then blends that distribution with the text model's probabilities. The influence of history increases with available prior-review count but is also subject to learned reliability gating, with a hard **maximum weight of 20%**. When no valid history exists, the mixture automatically falls back to text alone. This is a **20% cap, not a constant 20% contribution**.

V4 validation result	V2 text only	V4 text + history
Micro-F1 / accuracy	72.1905%	72.2032%
Macro-F1	57.2354%	57.2569%
MAE	0.370099	0.370004
Training time	20.12 min	20.59 min
Selected history weight cap	-	0.20 (single trial)

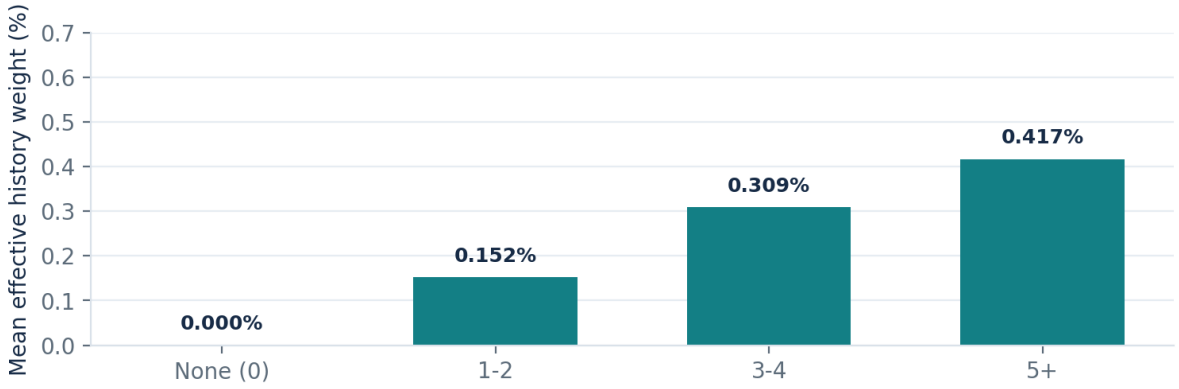


Figure 4. Mean learned history contribution by number of prior reviews on validation. Even for 5+ previous ratings, average contribution is only 0.417%.

What the small gain means

V4 exceeded V2 by only **0.0127 percentage points** in validation Micro-F1: approximately eight extra correct predictions out of 62,921. Macro-F1 rose by 0.0215 percentage points. The observed improvement is too small to establish a dependable history effect without repeated seeds or uncertainty analysis. The history gate effectively relied on text almost entirely; in the test run, mean history weight was just **0.0638%**.

Availability caveat: only 31.95% of V4 validation rows and 31.74% of its internal-test rows had eligible earlier reviews within the smaller modelling train bank. The full external audit reported 50.22% coverage under its different, larger history bank.

Source: amazon_bert_mini_V4_history_alpha(1)(2).ipynb; V2-to-V4 paired validation output, group-level alpha audit and training summary. Only alpha cap 0.2 was actually trained in this run.

CHAPTER 06 / EXPERIMENTAL REVIEW

Cross-version scorecard and learning dynamics

Version	Validation Micro-F1	Macro-F1	MAE	Batch	Text limit
V1	70.539%	54.706%	0.4122	64	128
V2	72.191%	57.235%	0.3701	128	256
V3	71.814%	56.688%	0.3765	128	256
V4	72.203%	57.257%	0.3700	128	256

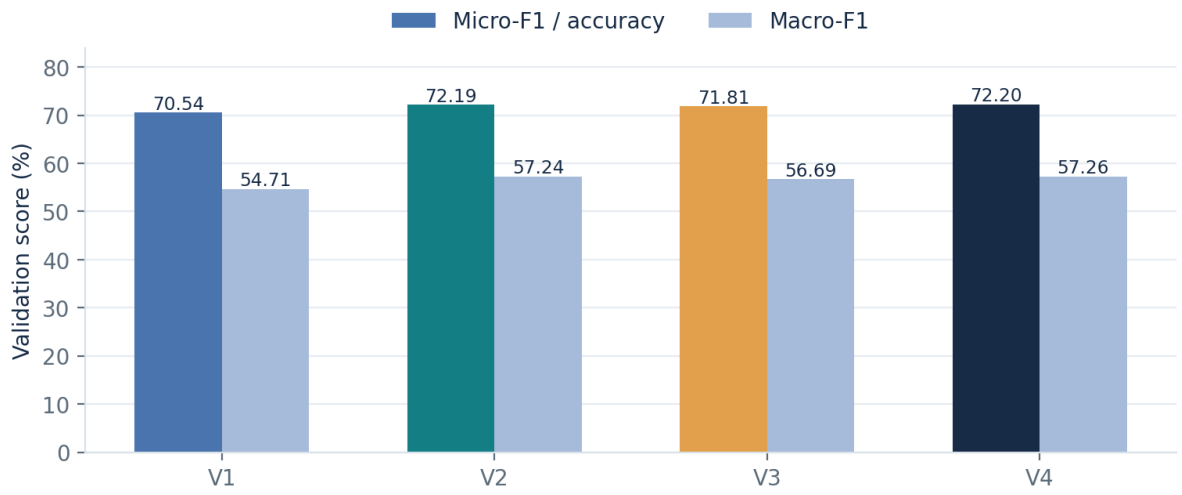


Figure 5. All four scores are on the validation split (n = 62,921), never on the external blind competition test.

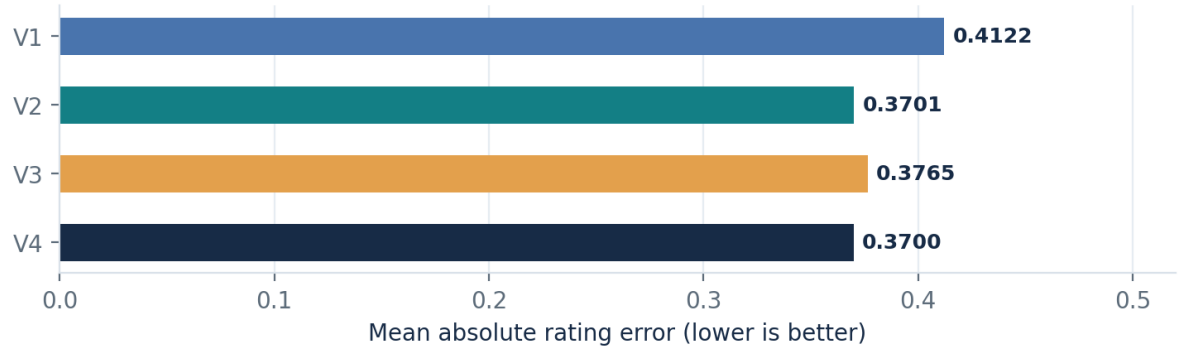


Figure 6. MAE improvement confirms fewer absolute star-rating errors for V2 and V4 (lower is better).

Learning curves and training efficiency

Performance through training

For V1, V2 and V4, saved per-epoch logs show Micro-F1 rising through epoch 4, which is also the best recorded epoch. These runs do **not** demonstrate that more epochs would improve generalization: later gains may saturate or reverse, and the schedule would have to be retuned. V3 has an overall four-epoch result, but its per-epoch numeric table was not preserved in the supplied notebook output.

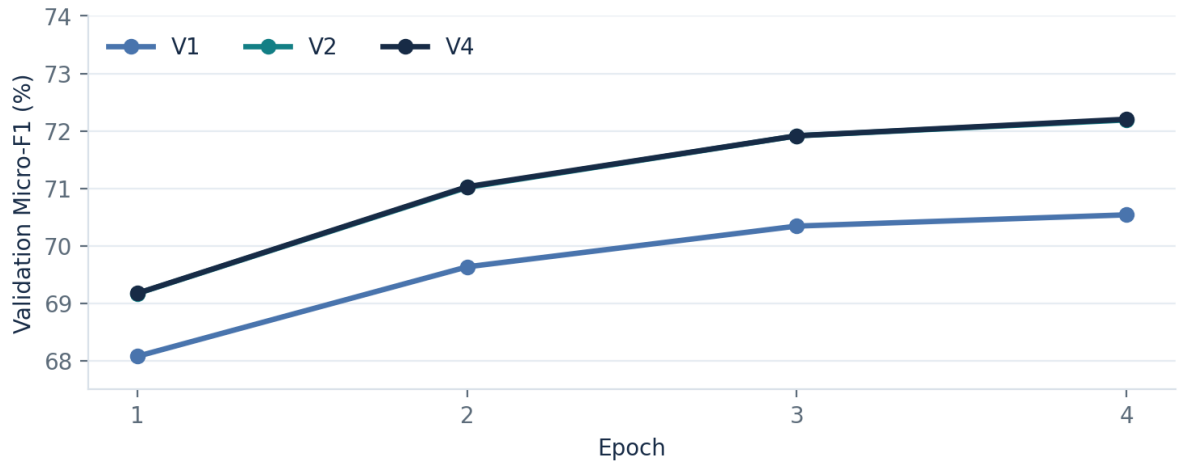


Figure 7. Recorded validation Micro-F1 after epochs 1-4 for the three versions with available per-epoch tables.

Training-time comparison

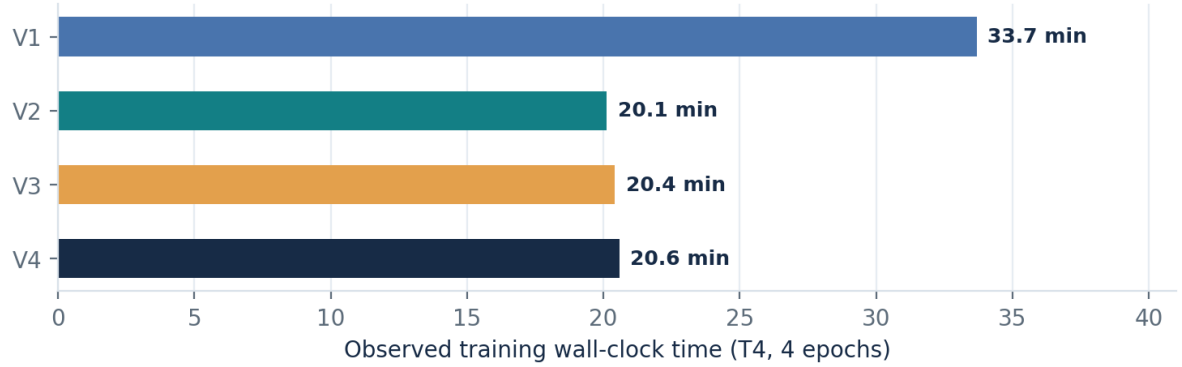


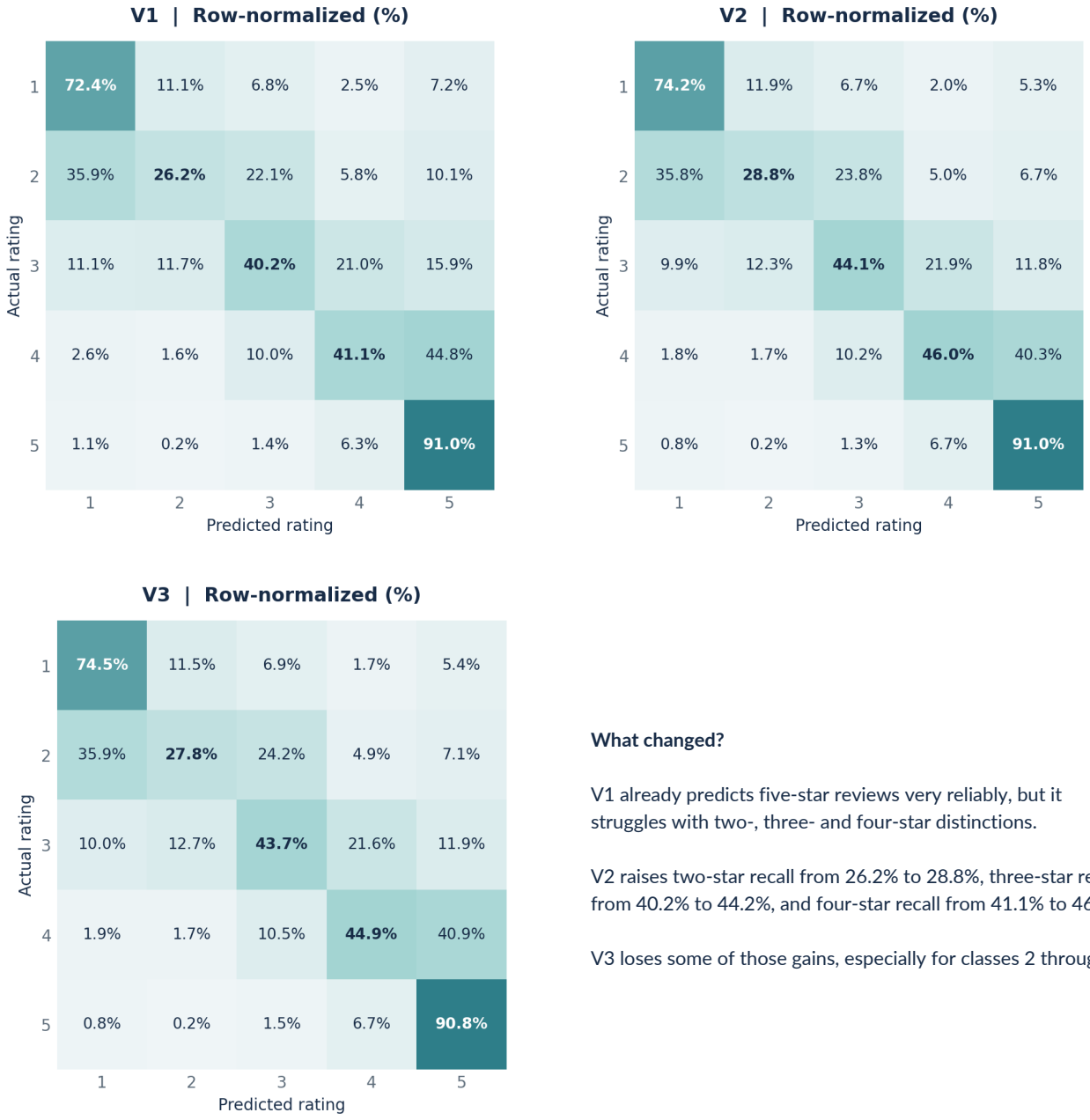
Figure 8. T4 wall-clock training only. V2/V3/V4 are each ~20 minutes; V1 is ~34 minutes. Token preprocessing and dataset audit time are separate from these training figures.

Interpretive hierarchy: **V2 - material improvement**; **V3 - rejected preprocessing intervention**; **V4 - marginal measured gain, uncertain incremental value**.

CHAPTER 07 / EXPERIMENTAL REVIEW

Confusion-matrix diagnostics on validation

The three validation confusion matrices below use identical actual/predicted star labels. Each row is normalized to 100%; darker diagonal cells indicate stronger class-specific recall. These are **validation results**, not the final test result. The V4 validation confusion matrix is not printed or embedded in the supplied V4 notebook.



What changed?

V1 already predicts five-star reviews very reliably, but it struggles with two-, three- and four-star distinctions.

V2 raises two-star recall from 26.2% to 28.8%, three-star recall from 40.2% to 44.2%, and four-star recall from 41.1% to 46.0%.

V3 loses some of those gains, especially for classes 2 through 4.

Figure 9. V1, V2 and V3 row-normalized validation confusion matrices. In all three, the main difficulty is the ambiguous middle of the star scale; five-star recall is above 90%.

A recurring error pattern

Four-star comments are often predicted as five stars: 5,257 cases in V1, 4,728 in V2 and 4,804 in V3. V2 reduces this particular error count despite unchanged class support, consistent with improved separation of moderately positive and strongly positive text.

Sources: Quera_V1(3).ipynb, Quera_V2(3).ipynb and Quera_V3(3).ipynb; full validation confusion-matrix tables embedded in their output cells.

CHAPTER 08 / EXPERIMENTAL REVIEW

Final outcome - saved V4 on internal held-out test

After selecting V4 using validation, the saved checkpoint was reloaded and evaluated **without any additional training** on the separate 62,921-row internal test partition (seed 42). Split disjointness was checked, and user history was recomputed from eligible earlier observations in the saved training bank.

72.057%
TEST MICRO-F1 / ACCURACY

57.069%
TEST MACRO-F1

0.3680
MEAN ABSOLUTE ERROR

62,921
INTERNAL TEST EXAMPLES

Final internal-test metric	Value
Micro-F1 / accuracy	0.720570
Macro-F1	0.570692
Weighted-F1	0.709290
MAE / NLL loss	0.368049 / 0.693352
Average effective history weight	0.000638 (0.0638%)
Inference time on T4	55.9 seconds

Detailed test errors

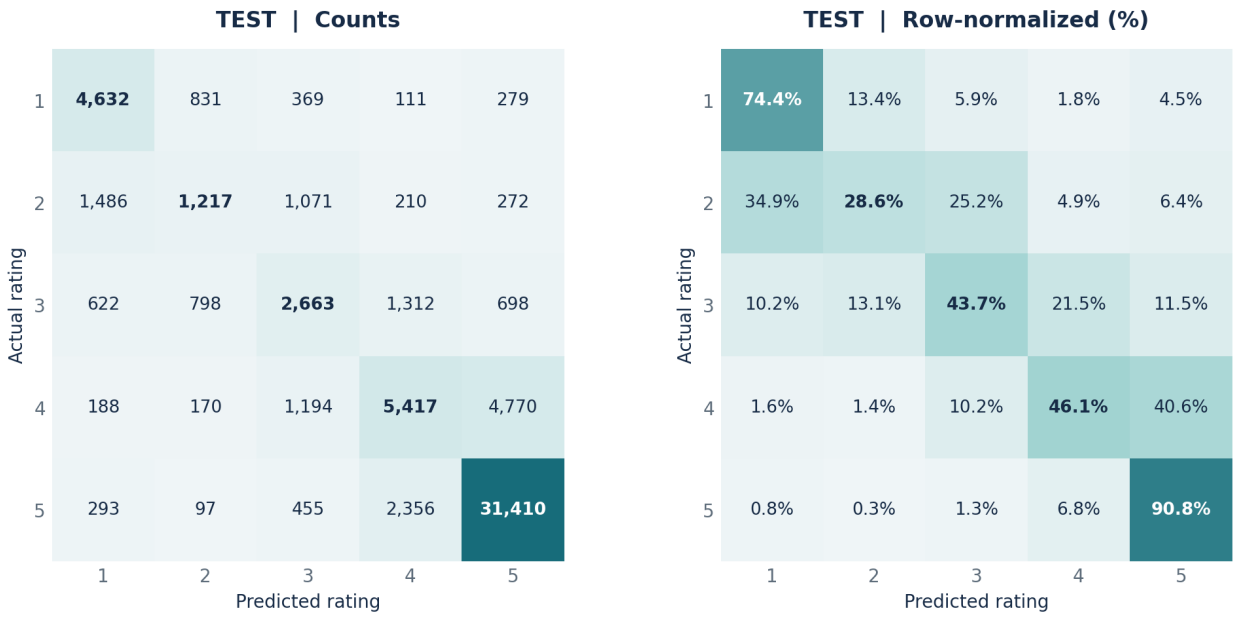


Figure 10. Final V4 internal-test confusion matrix, left: raw counts; right: within-actual-class percentages. Rows are actual rating, columns are predicted rating.

The 5-star class has **90.75% recall** and **87.20% F1**; 2-star reviews have only **28.59% recall** and **33.03% F1**. The largest recurring directional error is predicting four stars as five stars (4,770 cases). This is compatible with high overall Micro-F1 but a substantially lower Macro-F1.

Important: this is an **internal labelled holdout derived from the original train_data.csv**, not the **20,000-row external unlabelled test_data.csv**. No external competition score can be inferred from these notebooks.

Source: V4_History_Final_TEST_No_Training(1)(1).ipynb; logged sample counts, class report, five-by-five confusion matrix, history-group audit and final test metrics.

CHAPTER 09 / EXPERIMENTAL REVIEW

Conclusions, limitations and next decisions

What the sequence of experiments established

Decision	Assessment supported by measured results
Begin with compact pretrained BERT	Sound practical baseline; 70.54% validation Micro-F1 on half-data training
Increase meaningful context (V2)	Most impactful observed change; +1.65 pp Micro-F1 and +2.53 pp Macro-F1 vs V1
Filter stopwords (V3)	Did not help; token savings came with weaker prediction metrics
Add prior-user ratings (V4)	Feasible for many users; observed advantage vs V2 is only +0.013 pp Micro-F1
Test once after selection	Held-out internal test returned 72.06% Micro-F1 for the saved V4 checkpoint

Limitations that matter for interpretation

Single-seed evidence: each reported design has one visible executed run and no confidence interval; V4's extremely small advantage is not statistically established. **Bundled V2 changes:** sequence length, summary policy and batch size changed together; causal attribution requires ablations. **History exposure:** external blind-test history coverage (50.2%) was measured on the full labelled train, while V4's internal model used a half-data training bank (~31.7% history availability on internal test).

Non-identical review duplicates: the split audit identifies 1,732 distinct identical review texts shared between train and validation and 1,714 between train and internal test. Row identities are separate, but identical text may make the reported random-split performance optimistic relative to a group-aware or chronologically held-out evaluation. **Other constraints:** the task is class-imbalanced; final training was capped at four epochs, and the external 20,000-row blind test does not include rating labels.

Recommended follow-up experiments

Preserve V2 as the strong, simpler text-only reference. To establish incremental value of history, use repeated seeds and paired comparisons, quantify improvement specifically for history-rich users, and evaluate on a chronological or reviewer-group-aware holdout. Separately ablate max token length, summary usage, head-tail selection and batch size. Expand to the full labelled training set only after these design questions are resolved, keeping the external blind test untouched until a final submission.

Evidence index

Notebook	Evidence used
Quera_V1(3).ipynb	Baseline design, batch size, epoch logs, validation metrics / matrix
Quera_V2(3).ipynb	Summary + head-tail policy, batch 128, validation metrics / matrix
Quera_V3(3).ipynb	Stopword policy, token savings, validation metrics / matrix
test_train_user_comparison.ipynb	Full-train versus external blind-test user and temporal history audit
amazon_bert_mini_V4_history_alpha(1)(2).ipynb	History model, same split, paired V2 comparison, gate and validation
V4_History_Final_TEST_No_Training(1)(1).ipynb	Saved-checkpoint internal test and final confusion matrix

Reporting rule: **validation metrics compare designs; the internal-test metrics estimate the selected V4 model's held-out performance.** The 20,000-row external blind-test file has not been scored in any supplied output.